CSCI 645 Homework 2

Imitation Learning for Robot Manipulation: Demonstration Quantity and Quality for ACT on `PandaPickCube`

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Revised submission (protocol v2)

Environment	`PandaPickCube` (MuJoCo Playground), state-based, `impl=jax`
Teacher	PPO (Brax), 60,000,000 requested environment steps, seed 0
Student	ACT, chunk $H = 30$; two inference modes (ensembling / replan every step)
Experiments	Option A (quantity 25/50/100) and Option B (teacher action noise $\sigma \in \{0.5, 1, 2\}$)
Models	7 conditions $\times$ 3 training seeds = 21 students
Evaluation	suite A: 100 held-out ICs; suite B: 100 fresh ICs, frozen protocol
Hardware	NVIDIA A100-SXM4-80GB (dcwipphhgc216)

Summary

What was run. A PPO teacher on `PandaPickCube`; 100 clean demonstrations (baseline) plus nested 25/50 subsets (Option A) and 100 demonstrations collected with teacher action noise $\sigma \in \{0.5, 1, 2\}$ (Option B); 21 ACT/flow-matching students; every student evaluated under two inference modes on the same 100 held-out initial conditions (ICs; suite A), and the moderate-noise comparison re-tested on 100 *fresh* ICs under a protocol frozen and pushed before any outcome existed (suite B).

Findings. (1) Baseline ACT (100 clean demos) succeeds 24.3% of the time with temporal ensembling and 35.3% when replanning every step, against a teacher at 100%. (2) More clean demonstrations help in both modes (25 $\rightarrow$ 100: +13.7 and +26.3 percentage points). (3) Demonstrations from a *noisier* teacher ($\sigma = 0.5$, teacher success 70%) give a much *better* student when replanning (67.0% vs 35.3%, +31.7 pp), but not under ensembling (-3.7 pp, CI includes 0); $\sigma \geq 1$ destroys the task. (4) The frozen fresh-IC test **confirmed** the replanning result: 67.3% vs 20.3% ($\Delta = +47.0$ pp, IC 95% CI [+39.0, +54.7]). (5) The dominant baseline failures are not reaching *or* grasping alone: episodes split between never touching the box, touching it without lifting it, and lifting it without bringing it to the target.

Corrections in this revision. The termination-aware protocol changed *no* episode outcome in any of the 42 suite-A evaluations, so the original headline numbers were not inflated by post-termination states. The original failure analysis, however, was wrong: 14 of 27 successful baseline episodes had received failure labels (mostly "never reached the box"), because the classifier tested the 12 mm proximity before success. All checkpoints were retrained (the originals were lost); retrained per-seed results are close to the archived ones (Table 7).

Condition	#Demos	Teacher succ. (orig.)	Student, ensembling	Student, replan	Eval. episodes
ACT, 25 clean	25	100 (100)	10.7 [7.0, 14.7]	9.0 [5.3, 13.0]	3×100
ACT, 50 clean	50	100 (100)	14.0 [9.0, 19.7]	18.7 [13.0, 25.0]	3×100
ACT, 100 clean (baseline)	100	100 (100)	24.3 [18.0, 31.0]	35.3 [28.3, 42.3]	3×100
ACT, 100 at σ=0.5	100	65 (70)	20.7 [15.3, 26.3]	67.0 [59.7, 74.0]	3×100
ACT, 100 at σ=1	100	0 (1)	0.0 [0.0, 0.0]	1.0 [0.0, 2.3]	3×100
ACT, 100 at σ=2	100	0 (0)	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	3×100
Flow matching, 100 clean (suppl.)	100	100 (100)	24.3 [17.7, 31.3]	14.0 [9.3, 19.0]	3×100

Table 1: The table required by Part 4 (success in %, suite A, protocol v2). Teacher: the retrained teacher at the action-noise level of that condition's data on the same 100 ICs; in parentheses the original (data-collecting) teacher's archived v1 result. Student: mean of 3 training seeds with the IC-bootstrap 95% CI; each seed is evaluated on the same 100 held-out ICs (300 episodes per condition, not 300 independent ICs).

1 Task and demonstration source

Environment. `PandaPickCube` from MuJoCo Playground: a Franka Emika Panda must pick up a box and bring it to a target position sampled 0.2–0.4 m *above* the box's start (and within ± 0.2 m in x, y), so every success requires grasping and lifting. Observations are the 66-dimensional state vector of the environment's `_get_obs`: joint positions (16) and velocities (15), gripper position (3) and orientation (6), box orientation (6), box−gripper offset (3), target−box offset (3), target−box orientation difference (6) and the control residual (8). Actions are 8-dimensional joint-space deltas in $[-1, 1]$, scaled by `action_scale`= 0.04, added to the current control and clipped to the actuator limits. Episodes last 150 control steps of 0.02 s.

Defining success. The environment exposes no success signal: its `done` flag fires only when the box leaves the workspace ($|x|, |y|, |z| > 1$ m or $z < 0$) or the physics state contains NaN, and `state.info` only carries `reached_box` (gripper site within 12 mm of the box). We define:

> an episode succeeds if the box is within 5 cm of the target for 10 consecutive *valid* control steps (0.2 s), measured from the simulator's box and target positions.

The hold stops a fly-through from counting. Reaching the hold once is a success even if the box later leaves the radius; such later violations are reported separately.

Termination-aware evaluation (protocol v2). The raw environment has no auto-reset, so a batched rollout keeps stepping episodes whose box has already left the workspace. The original submission (protocol v1) scored success over the whole 150-step trace, including those post-termination states, and read the distance from the observation. Protocol v2, used for every number in this report, treats the first terminating transition's resulting state and all later states as invalid: they are excluded from success, distances, returns (only the terminating transition's own, finite, reward is counted) and every diagnostic; non-finite states are never valid; a hold cannot span termination, an invalid state or another episode. Unit tests cover normal success, an insufficient hold, an apparent success that occurs only after termination, a valid success before a later termination, NaN termination and cross-episode accumulation (`tests/test_metrics.py`). Per episode we save the reset keys, the valid length and terminal cause, the distance traces and every diagnostic, so all metrics can be recomputed without simulation. The observation-derived and simulator box–target distances agree to 1.2×10^{-7} m on valid states.

Teacher. PPO from Brax with the hyperparameters MuJoCo Playground ships for this task

σ	Orig. teacher	Retrained	Term. %	Dataset succ.	Data term.	$\lvert a_{\text{data}} - a_{\text{new}} \rvert$
0	100	100/100	0	100/100	0	0.351
0.1	100	99/100	0	99/100	0	0.337
0.3	98	97/100	0	97/100	0	0.415
0.5	70	65/100	10	71/100	7	0.516
1	1	0/100	7	0/100	13	0.678
2	0	0/100	7	0/100	10	0.809

Table 2: Teacher provenance. 'Orig. teacher': success (%) of the ORIGINAL teacher that collected the datasets (archived, protocol v1). 'Retrained': the retrained teacher on the same 100 held-out ICs (v2) and its share of terminated episodes. 'Dataset succ.': stored success labels of the committed 100-episode datasets (identical under v2). 'Data term.': dataset episodes truncated at termination. Last column: mean absolute difference between the stored actions and the retrained teacher's clipped action at the same states.

(`manipulation_params.brax_ppo_config`: 2048 parallel environments, learning rate 10^{-3}, discount 0.97, entropy cost 0.02, observation normalisation, policy MLP 4×32, value MLP 5×256), trained for 60,000,000 requested environment steps (brax ran 63,897,600) instead of the shipped 20 M, which was not enough for a useful demonstrator in the original run. Demonstrations use the *deterministic* policy action.

Provenance of the teacher and the data. The demonstration datasets in `data/` were collected by the *original* teacher, whose checkpoint was not recoverable. They are reused unchanged (SHA-256 recorded per model). The teacher was *retrained* with the same configuration and seed (11 minutes on one GPU); the reloaded pickle reproduces the in-memory policy (maximum action difference 0). The retrained teacher is a new artifact, *not* the policy that generated the data: on the stored clean states its action differs from the stored action by 0.351 on average (per-dimension absolute difference; actions lie in $[-1, 1]$). It is used only to report teacher success under protocol v2 and as the reference in one diagnostic. Table 2 separates the original teacher's (archived) success, the retrained teacher's success, and the success labels stored in the datasets. The initial state of every stored episode matches a reset from its recorded collection key (maximum difference 0).

Filtering. The clean dataset keeps successful episodes only, but the first 100 of the 512 collected all succeeded, so the filter removed nothing. The noisy datasets keep the first 100 episodes *including failures*, because data quality is the independent variable. All stored success labels are unchanged under protocol v2 (recomputed from the stored valid states; 0 disagreements in all six datasets).

2 Demonstration quality

The baseline (clean) dataset has 100 episodes and 15,000 transitions; every episode runs the full 150 steps (the environment never terminates on success) and all 100 succeed. The noisy datasets are shorter where the box was knocked out of the workspace (7, 13 and 10 episodes truncated at termination for $\sigma = 0.5, 1, 2$) and contain 71, 0 and 0 successful episodes. Figure 2 shows box–target distance traces, coverage of initial conditions and the executed-action distribution of the clean dataset; the expert video is `videos/expert_demo0_success.mp4` and Figure 4 shows frames.

Observations. The teacher is one deterministic policy, so trajectories from nearby initial conditions are highly consistent and unimodal; the traces show a reach phase, a lift that drives the distance down, and a hold. Initial box and target positions cover the sampled square uniformly by

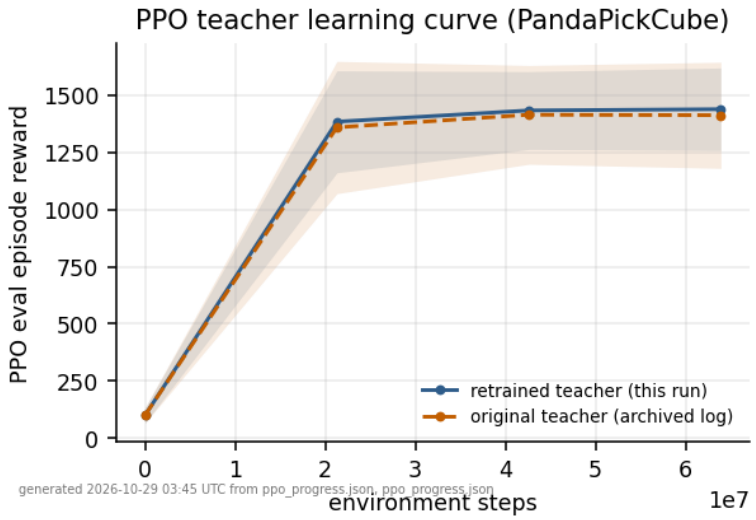


Figure 1: PPO teacher learning curves: the retrained teacher (this revision) and the original teacher (archived log). Bands: across-episode standard deviation at each evaluation point.

σ	Eps.	Transitions	Mean len.	Term.	Succ.	Filter	Used by
0	100	15,000	150.0	0	100	success only	quantity + flow
0.1	100	15,000	150.0	0	99	none	not used (preliminary)
0.3	100	15,000	150.0	0	97	none	not used (preliminary)
0.5	100	14,636	146.4	7	71	none	noise
1	100	14,093	140.9	13	0	none	noise
2	100	14,390	143.9	10	0	none	noise

Table 3: Committed demonstration datasets (collected by the original teacher). The clean set keeps successful episodes only (the first 100 of 512 collected all succeeded, so the filter removed none); noisy sets keep the first 100 episodes including failures. σ=0.1, 0.3 were a preliminary sweep that barely degraded the teacher and are not used by any student.

construction (100 draws). One environment quirk matters for the analysis: the reset observation is produced before MJX computes body positions, so in the first stored observation of every episode the gripper and box positions read as zero (later observations are correct; the student sees the same quirk at test time). Plots and all position-based diagnostics therefore start at the first stepped state. Visible inefficiencies: the approach to the box is often curved rather than direct, and near the target the box keeps moving slightly instead of settling, which is what the hold requirement measures.

3 Policy and training

We use **ACT** (action chunking with transformers). Input: the standardised 66-dimensional state. Output: a chunk of $H = 30$ future actions (8 each, i.e. 0.6 s). Architecture: the transformer CVAE of the ACT paper with hidden width 256, feed-forward width 1024, 8 heads, 4 encoder and 6 decoder layers, a 4-layer CVAE posterior encoder and a 32-dimensional latent (12.7 M parameters). The posterior encoder sees the observation and the ground-truth chunk and emits z; the decoder maps (z, obs) through H learned queries to the chunk. Objective: masked L_1 on the chunk plus

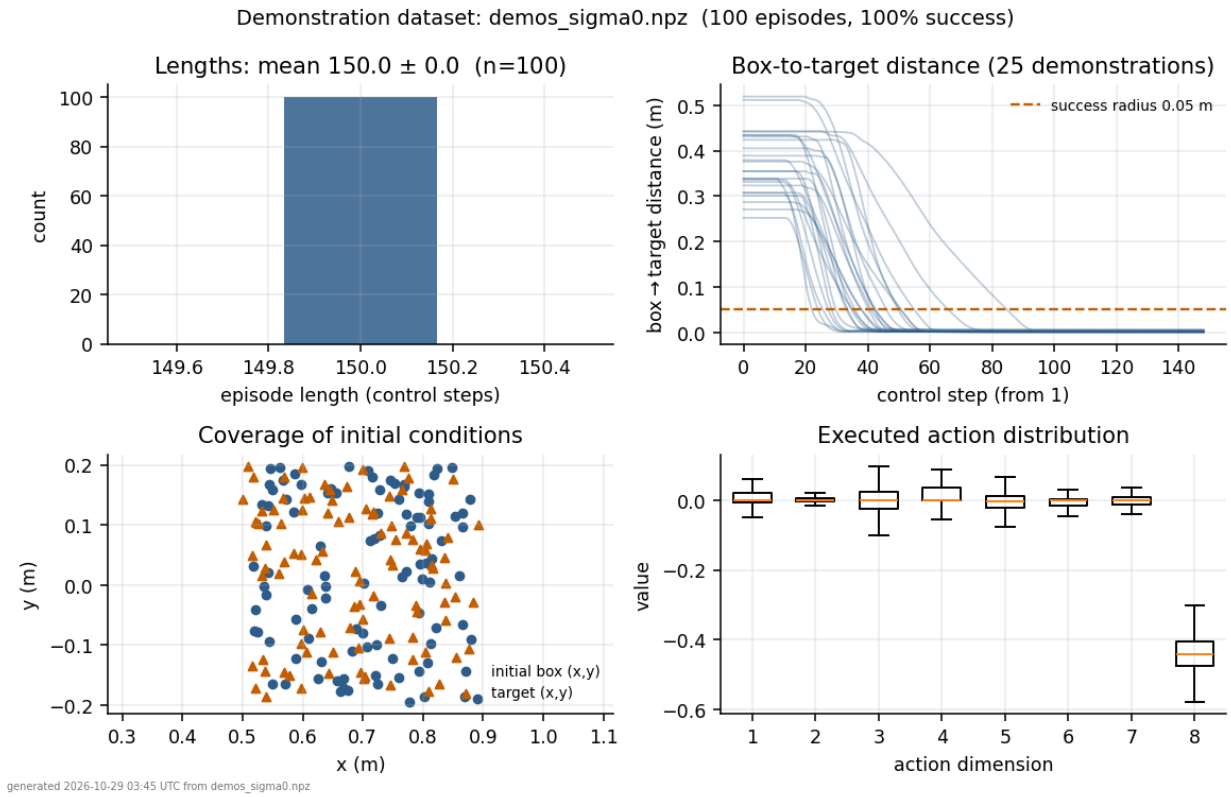


Figure 2: Clean demonstration dataset: episode lengths, box–target distance traces, initial box and target positions, and executed action distribution.

$10 \cdot \mathrm{KL}$; at test time $z = 0$. Chunks never cross an episode end (padded steps are masked).

Deviations from the reference implementation: no image backbone (state input, linear projection), pre-layer-norm GELU blocks, AdamW with warmup and cosine decay, bf16 autocast, gradient clipping and an EMA of the weights, which is what we evaluate. The code (`code/hw2/policies/act.py`) is a re-implementation; nothing is vendored.

Training. 6,000 steps, batch 128, learning rate 10^{-4} with 200 warmup steps and cosine decay to 5%, weight decay 10^{-4}, AdamW $\beta = (0.9, 0.95)$, gradient-norm clip 1, EMA decay 0.999. Observations and actions are standardised per dimension with statistics fitted on exactly the episodes each model trains on. Training seeds $\{0, 1, 2\}$ set initialisation, data order and CVAE sampling. One ACT model took 4.3 minutes on an A100 (eight models trained concurrently on four GPUs); a flow-matching model 1.5 minutes. Figure 3 shows all 21 loss curves.

Inference and action execution. Predicted chunks are de-standardised and clipped to $[-1, 1]$. Two modes are evaluated for every model: *temporal ensembling* (the historical headline protocol: the policy is queried every step and all chunk entries for the current step are averaged with weights $\exp(-0.01 \cdot \mathrm{age})$, so the newest prediction weighs most; ACT's reference weights the oldest most, which we evaluate as a diagnostic) and *replan every step* (a new chunk every step, only its first action executed). Neither weighting, normalisation, clipping nor the success threshold was changed relative to the original submission.

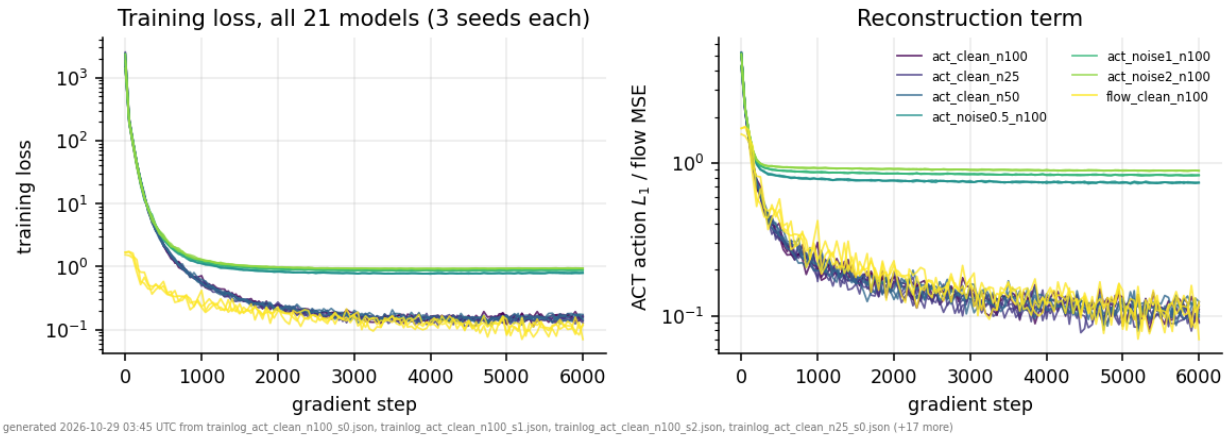


Figure 3: Training loss of all 21 models (ACT: total loss and action L_1; flow matching: velocity MSE).

4 Baseline results

Table 4 lists every model. The baseline (ACT, 100 clean demonstrations) succeeds on 24.3% of the 100 held-out ICs with temporal ensembling (seeds 26, 19, 28%; IC 95% CI [18.0, 31.0]) and 35.3% when replanning every step (seeds 35, 36, 35%; [28.3, 42.3]). The retrained teacher succeeds on 100/100 of the same ICs. The evaluation ICs (`split(PRNGKey(7777),100)`) are disjoint from every demonstration-collection key; no training trajectory is replayed. Rollout videos (verified as described in Section 6): `videos/student_act_clean_n100_s0_replan1_ep3_success.mp4` and `..._ep8_failure_lifted-the-box-never-within-target-radius.mp4`; Figure 4 shows frames.

Uncertainty. Each condition has three trained models, and all of them are evaluated on the same 100 ICs, so the 300 episodes are not 300 independent trials: some ICs are hard for every seed. We therefore report the mean of the three per-seed rates with two intervals: the *IC interval* resamples ICs with the trained models fixed (uncertainty about the IC distribution, conditional on these models), and the *2-way interval* also resamples training seeds. With three seeds the latter is crude, and neither interval describes the full distribution over training seeds; the per-seed values are always shown. Contrasts between conditions are paired by IC; p-values are sign-flip tests of the per-IC difference of seed-averaged success, conditional on the models.

Inference mode. Replanning every step beats ensembling for the baseline by +11.0 pp (IC CI [+5.3, +16.7]). Both modes were fixed in advance and are reported for every model; ensembling remains the historical headline mode. Executing chunks open loop is much worse (Table 9).

5 Data ablation

Option A: quantity. Students trained on nested prefixes ($25 \subset 50 \subset 100$) of the same clean dataset, with normalisation fitted on each subset; nothing else changes. Success rises with quantity in both modes (Figure 5, Table 5): ensembling 10.7 → 14.0 → 24.3%, replanning 9.0 → 18.7 → 35.3%. 25 → 100 is +13.7 pp [+7.0, +20.7] and +26.3 pp [+19.0, +33.7]; every seed improves. 25 → 50 is clear only when replanning (+9.7 pp vs +3.3 pp).

Option B: teacher action noise. Noise is added to the teacher's deterministic action *during*

Condition	Seeds 0/1/2	Mean	IC 95% CI	2-way CI	Seeds 0/1/2	Mean	IC 95% CI	2-way CI
ACT, 25 clean	11 / 5 / 16	**10.7**	[7.0, 14.7]	[4.3, 18.0]	11 / 7 / 9	**9.0**	[5.3, 13.0]	[4.7, 14.3]
ACT, 50 clean	15 / 13 / 14	**14.0**	[9.0, 19.7]	[8.3, 20.3]	19 / 19 / 18	**18.7**	[13.0, 25.0]	[12.3, 25.7]
ACT, 100 clean (baseline)	26 / 19 / 28	**24.3**	[18.0, 31.0]	[16.0, 33.0]	35 / 36 / 35	**35.3**	[28.3, 42.3]	[27.3, 43.3]
ACT, 100 at σ=0.5	19 / 25 / 18	**20.7**	[15.3, 26.3]	[13.7, 28.3]	70 / 67 / 64	**67.0**	[59.7, 74.0]	[58.3, 75.3]
ACT, 100 at σ=1	0 / 0 / 0	**0.0**	[0.0, 0.0]	[0.0, 0.0]	1 / 1 / 1	**1.0**	[0.0, 2.3]	[0.0, 2.7]
ACT, 100 at σ=2	0 / 0 / 0	**0.0**	[0.0, 0.0]	[0.0, 0.0]	0 / 0 / 0	**0.0**	[0.0, 0.0]	[0.0, 0.0]
Flow matching, 100 clean (suppl.)	25 / 29 / 19	**24.3**	[17.7, 31.3]	[16.0, 33.3]	17 / 14 / 11	**14.0**	[9.3, 19.0]	[8.0, 20.3]

Table 4: Suite A: task success (%) of every trained student on the original 100 held-out initial conditions (ICs), corrected protocol v2. Left block: temporal ensembling (historical headline mode); right block: replanning every step. 'Mean' averages the three training seeds. 'IC 95% CI' resamples the shared ICs with the trained models fixed; '2-way CI' also resamples the training seeds (crude with three seeds).

Contrast (B−A)	Δ (pp)	IC CI	2-way CI	p	Δ (pp)	IC CI	2-way CI	p
25 → 100 clean demos	+13.7	[7.0, 20.7]	[3.0, 24.0]	1.8×10^{-4}	+26.3	[19.0, 33.7]	[17.7, 35.0]	$< 10^{-4}$
50 → 100 clean demos	+10.3	[4.0, 16.7]	[1.3, 19.0]	0.0024	+16.7	[10.3, 23.0]	[8.7, 24.7]	$< 10^{-4}$
25 → 50 clean demos	+3.3	[-1.7, 8.3]	[-5.0, 11.3]	0.25	+9.7	[4.0, 15.7]	[2.7, 17.0]	0.0022
clean → σ=0.5	-3.7	[-11.7, 4.0]	[-14.3, 7.0]	0.41	+31.7	[22.0, 41.0]	[20.3, 42.3]	$< 10^{-4}$
clean → σ=1	-24.3	[-31.0, -18.0]	[-33.0, -16.0]	$< 10^{-4}$	-34.3	[-41.0, -27.7]	[-42.0, -26.7]	$< 10^{-4}$
clean → σ=2	-24.3	[-31.0, -18.0]	[-33.0, -16.0]	$< 10^{-4}$	-35.3	[-42.3, -28.3]	[-43.3, -27.3]	$< 10^{-4}$
ACT → flow matching	+0.0	[-8.7, 8.7]	[-11.7, 11.7]	1	-21.3	[-29.0, -14.0]	[-30.7, -12.0]	$< 10^{-4}$

Table 5: Suite A paired contrasts in success (percentage points), first block temporal ensembling, second block replanning every step. Both arms are evaluated on the same 100 ICs, so the IC bootstrap resamples ICs jointly; p is a sign-flip test of the per-IC difference of seed-averaged success (conditional on the trained models). Multiple contrasts are reported without correction.

environment interaction, $a = \text{clip}(\pi(s) + \varepsilon, -1, 1)$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, before the environment's action scaling and actuator clipping; the executed (noisy, clipped) action is stored with the observation it was chosen from. Dataset size is fixed at 100 episodes and failures are kept. A preliminary sweep at $\sigma \in \{0.1, 0.3\}$ barely changed the original teacher (success 100% and 98%; the closed-loop delta-action policy corrects the noise at the next step), so the study uses $\sigma \in \{0.5, 1, 2\}$, where the original teacher succeeded 70%, 1% and 0% of the time (the retrained teacher: 65, 0 and 0/100). The datasets contain 71, 0 and 0 successful episodes.

The result depends on the inference mode. With ensembling, $\sigma = 0.5$ is indistinguishable from clean (20.7% vs 24.3%, -3.7 pp [-11.7, +4.0]). With replanning, the $\sigma = 0.5$ students reach 67.0% against 35.3% (+31.7 pp [+22.0, +41.0], all three seeds). At $\sigma \geq 1$ the demonstrations no longer contain a successful pick-and-place and the students fail (1.0% and 0.0%).

Fresh-initialization confirmation (suite B). Because the $\sigma = 0.5$ result was first seen on the same 100 ICs used throughout, we froze a confirmation protocol (`configs/fresh_protocol.json`, committed and pushed on 2026-10-29 before any evaluation): the six existing checkpoints (clean and $\sigma = 0.5$, seeds 0–2), replanning, 100 fresh ICs from `PRNGKey(17777)` (key hash `5e18af4383e3`; no collision with any demonstration or suite-A key, nearest IC 3.1 mm away in box/target position), a fixed analysis and the decision rule "confirmed iff the IC 95% CI of the contrast lies above 0". Result (Table 6, Figure 6): clean 20.3%, $\sigma = 0.5$ 67.3%, $\Delta = +47.0$ pp (IC CI [+39.0, +54.7], 2-way [+37.7, +56.0], sign-flip $p < 10^{-4}$; per seed +49, +47, +45 pp): the finding is **confirmed** on unseen ICs. For continuity, the majority-vote McNemar test (an IC counts as solved if ≥ 2 of 3 seeds solve it, a different statistic from the mean difference) gives 48 vs 2 discordant ICs, $p < 10^{-4}$.

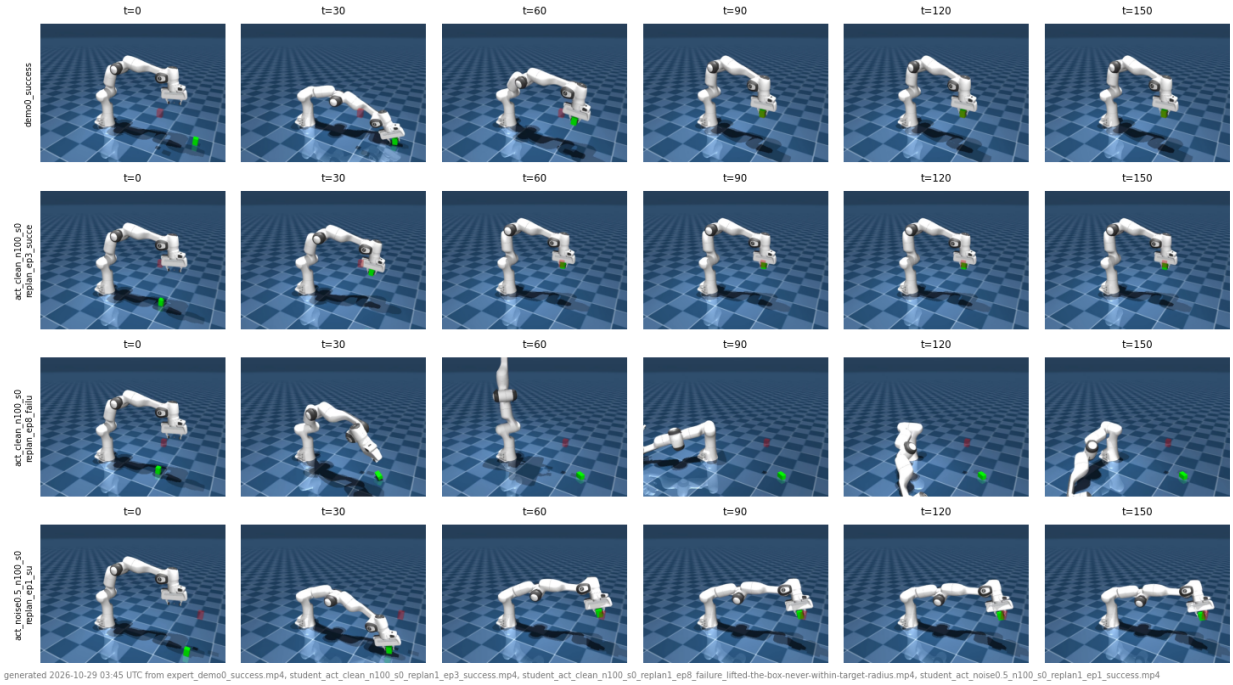


Figure 4: Frames from the expert demonstration and the four verified student clips (baseline and $\sigma = 0.5$, seed 0, replanning; one success and one failure each). Every 2nd control step is rendered; t is the control step.

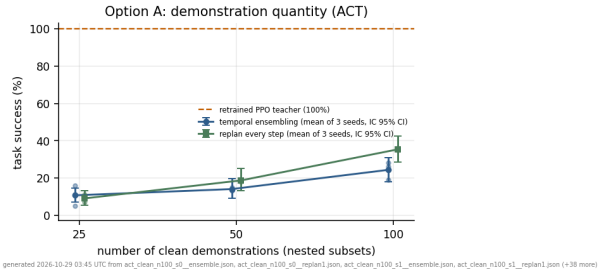
Suite B (replan)	Seed 0	Seed 1	Seed 2	Mean	IC 95% CI	2-way CI
ACT, 100 clean (baseline)	21/100	20/100	20/100	**20.3**	[14.7, 26.3]	[14.0, 27.3]
ACT, 100 at σ=0.5	70/100	67/100	65/100	**67.3**	[59.7, 74.7]	[58.7, 75.7]
σ=0.5 − clean	+49	+47	+45	**+47.0**	[+39.0, +54.7]	[+37.7, +56.0]

Table 6: Suite B, the frozen confirmation: the same six checkpoints on 100 fresh initial conditions (`PRNGKey(17777)`), replanning every step, 600 episodes. Last row: noisy minus clean, per seed index and for the mean (percentage points); sign-flip $p < 10^{-4}$.

The clean baseline scores clearly lower on the fresh ICs (20.3%) than on the original ones (35.3%; the IC intervals do not overlap), while the $\sigma = 0.5$ students score the same on both (67.3% vs 67.0%): the original 100 ICs were apparently easier than average for the clean students, which also means the suite-A baseline numbers are somewhat optimistic. Three training seeds limit what can be said about other seeds.

What the noise result does and does not show. Moderate noise changes two things at once: the actions are worse (the teacher at $\sigma = 0.5$ fails about a third of the time) and the visited states are more varied (perturbed trajectories, including recoveries from perturbations). We did not run an experiment that separates the two, e.g. noisy collection keeping only successful episodes, or noise injected during collection but the clean action stored as the label (DART-style). The result is consistent with the covariate-shift account in Section 6, but that is a hypothesis, not an isolated mechanism.

Supplementary: flow matching. A flow-matching policy (DiT/adaLN-Zero denoiser over



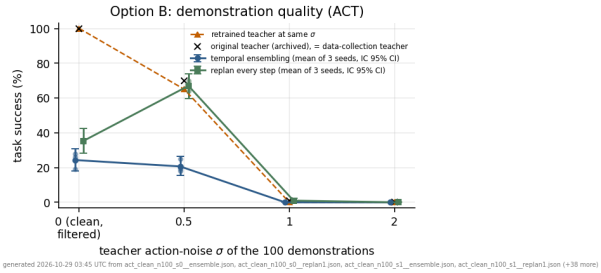
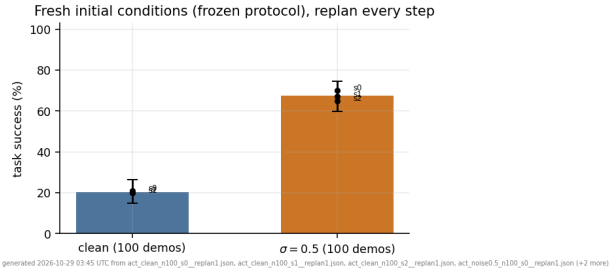


Figure 5: Option A (left) and Option B (right). Lines: mean of three seeds with IC 95% intervals; dots: individual seeds; both inference modes. Right: teacher success at each σ (retrained teacher, dashed; original teacher, crosses).



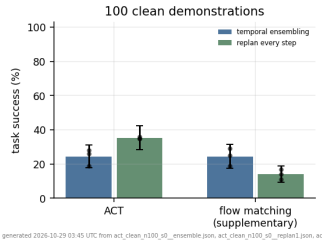


Figure 6: Left: suite B, the frozen fresh-IC confirmation (bars: mean, IC 95% CI; dots: seeds). Right: supplementary flow-matching arm versus ACT.

the action chunk, 5.1 M parameters) on the same 100 clean demos ties ACT under ensembling (24.3% vs 24.3%) and is worse when replanning (14.0%, -21.3 pp), plausibly because resampling a fresh stochastic chunk every step makes consecutive actions inconsistent. This arm is supplementary; the required baseline and both ablations use ACT only.

Historical versus corrected results. Table 7 compares the original submission's numbers (protocol v1, original checkpoints) with this revision (v2, retrained checkpoints). The two differ by a few points per condition, and the conclusions are unchanged. Because the checkpoints differ, those differences are not attributable to the evaluation fix. What *is* attributable to it is measured directly: re-scoring the new rollouts with the v1 rule changes zero outcomes in all 42 evaluations, because terminated episodes had never re-entered the success region after termination.

6 Failure analysis

Correcting the original analysis. The original analysis of the baseline (seed 0, ensembling) reported 27 successes but labelled only 13 episodes "success" and 66 "never reached the box", with only 18 episodes inside the 12 mm `reached_box` radius. Its classifier tested termination and the 12 mm threshold *before* success, so 14 successful episodes received failure labels. The 12 mm gripper-site-to-box-centre distance is also not evidence of reaching: among all 814 successful episodes in this revision (teacher and students), every one shows finger–box contact and lifts the box by at least 20.6 cm, yet only 76% come within 12 mm (median closest approach 8 mm, maximum 29 mm).

Taxonomy. Each episode now receives exactly one label, decided in a fixed order on valid states only: success; terminated early; never touched the box (no finger-pad/box contact in MJX's

Condition	v1 (orig.)	v2 (new)	flips	v1 (orig.)	v2 (new)	flips
ACT, 25 clean	11.3	10.7	0	7.7	9.0	0
ACT, 50 clean	14.7	14.0	0	19.3	18.7	0
ACT, 100 clean (baseline)	26.0	24.3	0	32.0	35.3	0
ACT, 100 at σ=0.5	20.7	20.7	0	66.7	67.0	0
ACT, 100 at σ=1	0.0	0.0	0	1.0	1.0	0
ACT, 100 at σ=2	0.0	0.0	0	0.0	0.0	0
Flow matching, 100 clean (suppl.)	22.0	24.3	0	12.0	14.0	0

Table 7: Historical (protocol v1, original checkpoints) versus corrected (v2, retrained checkpoints) mean success (%), ensembling then replanning. The checkpoints differ, so the difference is NOT attributable to the evaluation fix alone. 'flips' counts, on the new rollouts, the episodes (of 300) whose outcome changes between the v1 and v2 rules – the part that IS due to the fix.

Model	Mode	Succ.	Term.	No touch	No lift	Lifted, far	Short hold
ACT, 100 clean (baseline)	temporal ensembling	73	6	82	91	28	20
ACT, 100 clean (baseline)	replan every step	106	12	50	77	46	9
ACT, 100 at σ=0.5	temporal ensembling	62	11	11	99	100	17
ACT, 100 at σ=0.5	replan every step	201	3	3	42	50	1

Table 8: Mutually exclusive episode outcomes, pooled over 3 seeds $\times$ 100 ICs (suite A). Decision order: success; terminated early; never touched the box (no finger-pad contact); touched but never lifted it $\geq$2 cm; lifted but never within 5 cm of the target; within 5 cm but held <10 steps.

contact list); touched but never lifted it (box never ≥ 2 cm above its resting height); lifted but never within 5 cm of the target; within 5 cm but held < 10 steps. The thresholds were checked against teacher and student trajectories as above and against the videos. Counts sum to the number of failures (Table 8, Figure 7). For the baseline with ensembling, the 227 failures are: never touched 82, touched but not lifted 91, lifted but never near the target 28, too short a hold 20 and terminated 6. So the student fails at every stage: it often does not get the fingers onto the box, often does not achieve a grasp that lifts it, and sometimes lifts it but does not carry it to the target. The $\sigma = 0.5$ students with replanning almost never miss the box (3 of 300) and fail mainly after contact (42 not lifted, 50 lifted but far).

Diagnostics (separate runs, Table 9). (i) The student fits its data: the one-step action error on the stored training states is 0.031–0.032 across seeds (action units; actions lie in $[-1, 1]$). (ii) Evaluated against the retrained teacher, the error is 0.073 on states the teacher visits and 0.653 on states the student visits, about 9.0$\times$ larger. The reference here is the retrained teacher, which itself differs from the stored actions by 0.351 on the training states, so only the comparison between the two state distributions is meaningful, not the absolute values. A low average error on teacher states also does not rule out small, control-sensitive errors at critical moments such as closing the gripper. (iii) The student is only slightly better on the ICs its demonstrations came from than on held-out ICs (ensembling 38/27, 32/21, 27/28 successes per seed), so the gap to the teacher is not a matter of unseen initial conditions. (iv) Inference: executing whole chunks open loop is worst; ACT-reference ensemble weighting (oldest prediction weighted most) is no better than the historical weighting; replanning is best.

Interpretation. Together these point to compounding error under covariate shift, the standard behaviour-cloning failure: the policy is accurate where the teacher goes and drifts into states the demonstrations do not cover, where nothing teaches it to recover. Replanning every step limits how

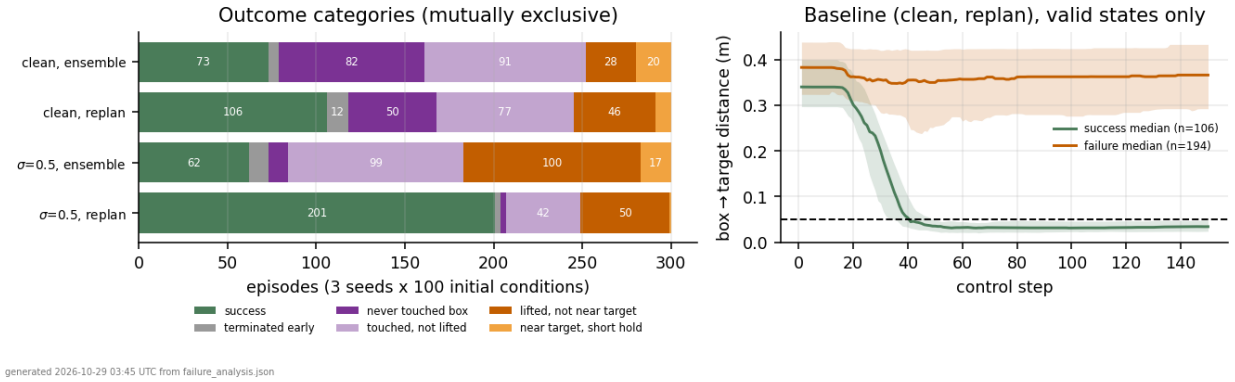


Figure 7: Left: outcome categories, pooled over three seeds × 100 ICs. Right: box–target distance of the baseline (replanning) over valid states, median and interquartile band, by outcome.

Baseline	Ens. train/test	Replan train/test	MAE data	MAE teacher	MAE student	Ens. (ACT order)	Open loop	Chunk 5
seed 0	38 / 27	38 / 35	0.031	0.073	0.653	26	10	25
seed 1	32 / 21	35 / 34	0.031	0.073	0.651	18	6	25
seed 2	27 / 28	33 / 31	0.032	0.075	0.662	28	9	23

Table 9: Separate diagnostic runs on the three baseline models (ACT, 100 clean). Columns 2–3: successes out of 100 on the ICs the demonstrations came from vs. the held-out ICs. MAE: mean absolute error of the first predicted action on the stored training states (vs. stored actions) and on states visited by the retrained teacher or by the student (vs. the retrained teacher; its own disagreement with the stored actions on training states is 0.351). Last three: successes of other inference variants.

long an error persists, and noisy demonstrations add perturbed and recovering states to the data, which fits both of our main effects. This is the best-supported explanation, not a proven one: the diagnostics show the error grows off the teacher's distribution, not that coverage is the only cause, and the noise ablation does not separate coverage from action quality.

Determinism and the videos. Batched GPU evaluation is not bit-reproducible: re-running the same 100-episode evaluation reproduces 96–97 of the 100 outcomes, so per-model success rates are reproducible to within a few episodes (the diagnostic re-runs in Table 9 show the same effect). The videos are therefore never named after a different run. Each clip comes from one batched 100-episode rollout, is labelled by that rollout's own outcome, and is accepted only if re-scoring the rendered states reproduces the label exactly.

7 Reproducibility and reflection

Software. Python 3.12.15, jax/jaxlib 0.9.2 (CUDA 12), flax 0.11.2, brax 0.14.2, mujoco and mujoco-mjx 3.15.0, `playground` 0.2.0 at commit `d59156e099a8` (assets: mujoco_menagerie `1b86ece57659`), torch 2.11.0+cu128, numpy 2.5.4. Hardware: NVIDIA A100-SXM4-80GB. Full pins: `configs/requirements.lock.txt`.

Commands. `README.md` lists every step: `setup_env.sh`; `stage1r_teacher.py` (teacher, teacher evaluation, dataset audit); `stage2_train.py` (21 models); `stage2_eval.py --suite A/B`; `stage2b_failure.py`, `diag_student.py`, `diag_replay.py`; `collect_run_artifacts.py`;

`stage3_report.py` (CPU only, builds this PDF from committed results); `package.py`. Paths derive from the checkout. Training and evaluation resume only when a stored identity (dataset, configuration, code, checkpoint and evaluation-key hashes, protocol version) matches. Checkpoints are attached to the repository as release assets with SHA-256 hashes (`results/checkpoints_manifest.json`). Seeds: students $\{0, 1, 2\}$; suite A `split(PRNGKey(7777),100)`; suite B `split(PRNGKey(17777),100)`; demonstrations `split(PRNGKey(1000+`$\lfloor 1000\sigma \rfloor$`),512)`; PPO seed 0.

Environment issues worth recording. (i) Our GPU nodes have no NVIDIA graphics userspace, so the assignment's EGL snippet cannot run there; videos are rendered with software OSMesa from a side conda prefix (`HW2_GL=osmesa`), configured before `mujoco` is imported because PyOpenGL resolves `libOSMesa` via `ctypes.util.find_library`, which ignores `LD_LIBRARY_PATH`. (ii) brax 0.14.2 calls `jax.device_put_replicated`, removed in newer jax, hence the jax/flax pins. (iii) MJX reproduces an initial condition exactly (stored versus re-reset observations agree to 0), but replaying a stored action sequence diverges chaotically at floating-point level (deviation $> 10^{-3}$ first at steps 27, 22, 36 in three episodes), because batched and single-environment rollouts do not run identical kernels. Videos are therefore rendered from the *same* batched rollout that is scored, not from a single-environment replay.

Runtime (measured). Teacher: 11 minutes of PPO on one A100. Students: 4.3 minutes per ACT model and 1.5 per flow model, eight at a time on four A100s (about 15 minutes for all 21). Evaluation: about 54 s per model, mode and 100 ICs. The CPU-only report rebuild takes about a minute plus LaTeX.

Limitations. Three training seeds per condition; one task; one success definition (5 cm, 10 steps); a retrained teacher that is not the data-collecting teacher, so teacher-referenced diagnostics are approximate; evaluation reproducible only to within a few episodes per model; and the noise ablation confounds action quality with state coverage.

With more data, compute or time. (i) Separate coverage from quality: inject noise during collection but store the clean teacher action (DART), and collect noisy data keeping only successful episodes. (ii) Attack covariate shift directly with DAgger-style relabelling by the teacher. (iii) More training seeds and a deterministic evaluation mode (deterministic XLA/cuDNN kernels), so per-model numbers reproduce exactly. (iv) Check the conclusions on a second manipulation task and under other success radii.

8 Citations and use of tools

- MuJoCo Playground: Zakka et al., *MuJoCo Playground*, 2025, https://arxiv.org/abs/2502.08844; code https://github.com/google-deepmind/mujoco_playground (commit `d59156e099a8`). Environment, PPO hyperparameters and the random-action snippet of the assignment.
- MuJoCo Menagerie (Panda model): https://github.com/google-deepmind/mujoco_menagerie.
- ACT: Zhao et al., *Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware*, RSS 2023, https://arxiv.org/abs/2304.13705; reference implementation https://github.com/tonyzhaozh/act. Re-implemented with the deviations of Section 3.
- Diffusion Policy (the alternative the assignment names; not used): Chi et al., RSS 2023,

Artifact	Produced from
results/summary.json, results/summary.csv	results/eval/A/*.json, results/eval/B/*.json
results/figures/ppo_curve.png	configs/teacher/ppo_progress.json, results/historical_v1/ppo_progress.json
results/figures/demos.png	data/demos_sigma0.npz
results/figures/loss.png	configs/trainlogs/*.json
results/figures/quantity.png	results/eval/A/*.json
results/figures/noise.png	results/eval/A/*.json, results/teacher/teacher_report.json
results/figures/algorithms.png	results/eval/A/*.json
results/figures/fresh.png	results/eval/B/*.json
results/figures/failure.png	results/failure_analysis.json, results/episodes/A/act_clean_n100_s*__replan1.npz
results/figures/frames.png	videos/*.mp4
results/tables/*.tex	results/summary.json, results/teacher/teacher_report.json, results/failure_analysis.json, results/diag/*.json
report/numbers.tex	all of the above, results/environment.json, configs/*

Table 10: Provenance of every figure and table (all built by `scripts/stage3_report.py`).

https://arxiv.org/abs/2303.04137.

- Flow matching (supplementary arm): Lipman et al., ICLR 2023, https://arxiv.org/abs/2210.02747; Liu et al., ICLR 2023, https://arxiv.org/abs/2209.03003; DiT/adaLN-Zero: Peebles and Xie, ICCV 2023, https://arxiv.org/abs/2212.09748.
- Brax (PPO): Freeman et al., https://github.com/google/brax.
- Covariate shift in behaviour cloning: Ross, Gordon and Bagnell, *A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning* (DAgger), AISTATS 2011; noise injection during demonstration: Laskey et al., *DART*, CoRL 2017, https://arxiv.org/abs/1703.09327.
- **LLM use.** An LLM coding assistant (Anthropic's Claude, in Claude Code) wrote and debugged most of the implementation, the cluster job scripts and the evaluation and statistics code, diagnosed the OSMesa and jax/brax environment problems, and drafted this report, in both the original submission and this revision. In the revision it also audited the original code, found the termination and failure-labelling errors described in Section 6, implemented the fixes and tests, retrained and re-evaluated all models, and rewrote the report. Every number in this report is generated by `stage3_report.py` from stored result files (each quantity in the prose is a LaTeX macro, not typed by hand). The author specified the assignment scope and the revision requirements and is responsible for the submission.